Green University of Bangladesh
Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering
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LAB REPORT NO: 02

Course Title : Design Project II
Course Code : CSE 320 Section : 231_D1

Student Details

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Lab Date : 24-02-26
Submission Date : 03-03-26
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[For Teachers use only: Don't Write Anything inside this box]

Lab Report Status

Marks: …… **Signature:.....**
Comments:..... **Date:.....**

1. TITLE :

Development of Backend Infrastructure: MySQL Database Integration and PHP Server-Side Logic for the **EduAtlas** Platform**.**

2. OBJECTIVES :

- To design and initialize a relational database schema using MySQL.
- To facilitate secure communication between PHP scripts and the database engine via PDO.
- To develop core server-side modules for user interaction.
- To engineer a robust authentication system featuring session management and role-based access control (Student vs. Admin).

3. PROCEDURE :

The primary focus of this lab session was the architectural setup of EduAtlas, a comprehensive web portal designed to streamline the academic admission landscape in Bangladesh. The development followed these milestones:

Create database:

- Created database named **`eduatlas`**.
- Designed tables for:
  - **Users :** Stores student and admin login information
  - **User_profiles :** Stores GPA, group, preferred location,and target level.
  - **Institutions :** Stores college, university, and abroad institution information
- Inserted a default admin account into the users table.

Connect PHP with database:

- Stored DB settings in **`app/config.php`**
- Created PDO connection in **`app/db.php`**
- Used a helper function **`db()`** to reuse the same connection.

Start server-side operations:

- Created homepage using **index.php**.
- Implemented student login (session-based) using **login.php**.
- Implemented admin login using **admin_login.php**.
- implemented dynamic **navigation bar (navbar)** for the user panel.

4. IMPLEMENTATION & OUTPUT :

Database Configuration :

```
<?php

return [

  'base_url' => '/EduAtlas',

  'db' => [
    'host' => '127.0.0.1',
    'port' => '3306',
    'name' => 'eduatlas',
    'user' => 'root',
    'pass' => '',
    'charset' => 'utf8mb4',
  ],
  'session_name' => 'EDUATLASSESSID',
];
```

Database Connection using PDO :

```
function db(): PDO {
  static $pdo = null;
  if ($pdo) return $pdo;

  $c = (require __DIR__ . '/config.php')['db'];
  $dsn =
"mysql:host={$c['host']};port={$c['port']};dbname={$c['name']};charset={
$c['charset']}";

  $pdo = new PDO($dsn, $c['user'], $c['pass'], [
    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
  ]);

  return $pdo;
}
```

Database of EduAtlas :

```
CREATE DATABASE eduatlas;
USE eduatlas;

CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('student','admin') NOT NULL DEFAULT 'student',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE student_profiles (
  user_id INT PRIMARY KEY,
  ssc_gpa DECIMAL(3,2) NULL,
  hsc_gpa DECIMAL(3,2) NULL,
  grp ENUM('Science','Business','Arts') NULL,
  budget_min INT NULL,
  budget_max INT NULL,
  preferred_location VARCHAR(120) NULL,
  target_level ENUM('College','University') NULL,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE institutions (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(200) NOT NULL,
  type ENUM('College','University','Abroad') NOT NULL,
  location VARCHAR(120) NOT NULL,
  is_public TINYINT(1) NOT NULL DEFAULT 0,
  min_gpa DECIMAL(3,2) NULL,
  budget_min INT NULL,
  budget_max INT NULL,
  admission_requirements TEXT NULL,
  official_website VARCHAR(255) NULL,
  description TEXT NULL,
  image_url VARCHAR(255) NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;


-- Default admin:

INSERT IGNORE INTO users (id, name, email, password_hash, role)
VALUES (1, 'Admin', 'admin@eduatlas.local',
'$2y$10$2vB3pUu7Wm8uJ8iU9nEJleq5I6xgQfJqkqB0zTn8Hq2qkXGgJz', 'admin');
```

Start / Home Page:

```
require_once __DIR___ . '/includes/bootstrap.php';

$title = 'EduAtlas - Home';
require __DIR___ . '/includes/header.php';
require __DIR___ . '/includes/navbar_public.php';
```

User Login :

```
if ($_SERVER['REQUEST_METHOD'] === 'POST') {

  $email = trim($_POST['email'] ?? '');
  $password = (string)($_POST['password'] ?? '');

  $st = db()->prepare("SELECT * FROM users WHERE email=? LIMIT 1");
  $st->execute([$email]);
  $u = $st->fetch();

  if ($u && password_verify($password, $u['password_hash'])) {

    $_SESSION['user'] = [
      'id' => (int)$u['id'],
      'name' => $u['name'],
      'email' => $u['email'],
      'role' => $u['role'],
    ];

    header("Location: " . base_url('/student_dashboard.php'));
    exit;
  }
}
```

Admin Login :

```
if ($_SERVER['REQUEST_METHOD'] === 'POST') {

  $email = trim($_POST['email'] ?? '');
  $password = (string)($_POST['password'] ?? '');

  $st = db()->prepare("SELECT * FROM admins WHERE email=? LIMIT 1");
  $st->execute([$email]);
  $a = $st->fetch();

  if ($a && password_verify($password, $a['password_hash'])) {
    $_SESSION['user'] = [
      'id' => (int)$a['id'],
      'name' => $a['name'],
      'email' => $a['email'],
      'role' => 'admin',
    ];
  }
}
```

Navigation bar (navbar) for the user :

```
    <nav class="px-4 mt-2 space-y-2">
      <?= nav_item('student_dashboard.php', 'Dashboard', $icoDashboard,
is_active('student_dashboard.php')) ?>
      <?= nav_item('student_discover.php', 'Discover', $icoDiscover,
is_active('student_discover.php')) ?>
      <?= nav_item('student_bookmarks.php', 'Bookmarks', $icoBookmark,
is_active('student_bookmarks.php')) ?>
      <?= nav_item('student_admission_help.php', 'Admission Help',
$icoHelp, is_active('student_admission_help.php')) ?>
      <?= nav_item('student_study_materials.php', 'Study Materials',
$icoStudy, is_active('student_study_materials.php')) ?>
      <?= nav_item('student_resources.php', 'Resources', $icoRes,
is_active('student_resources.php')) ?>
      <?= nav_item('student_about.php', 'About', $icoAbout,
is_active('student_about.php')) ?>
    </nav>
```

5. OUTPUT :

Figure 01 : Database of **EduAtlas** (Some tables are still being added).

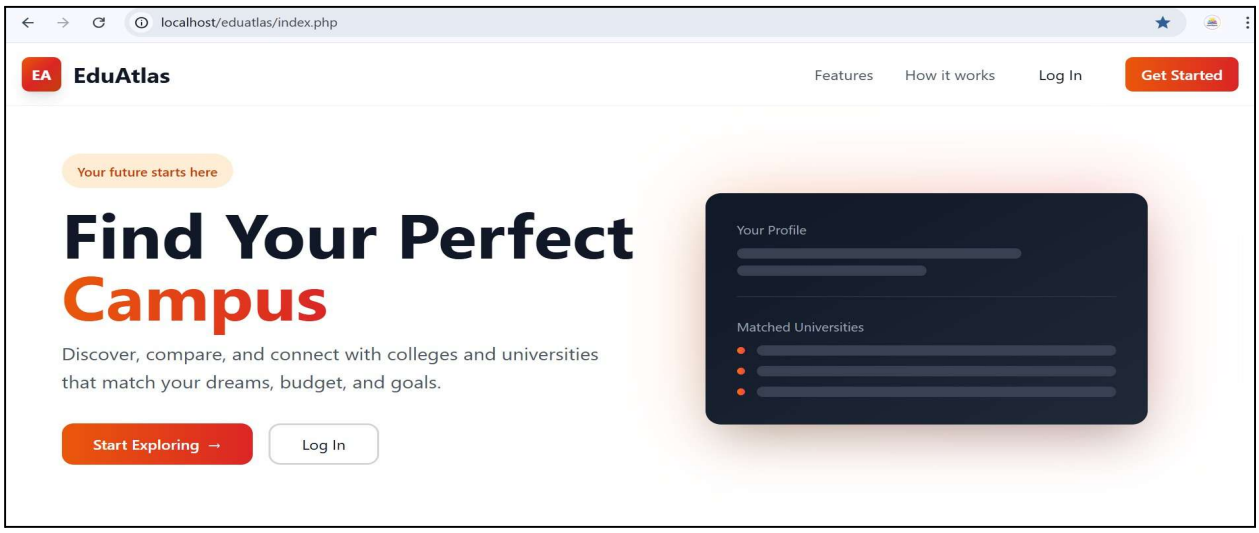

Figure 02 : Home page of **EduAtlas.**

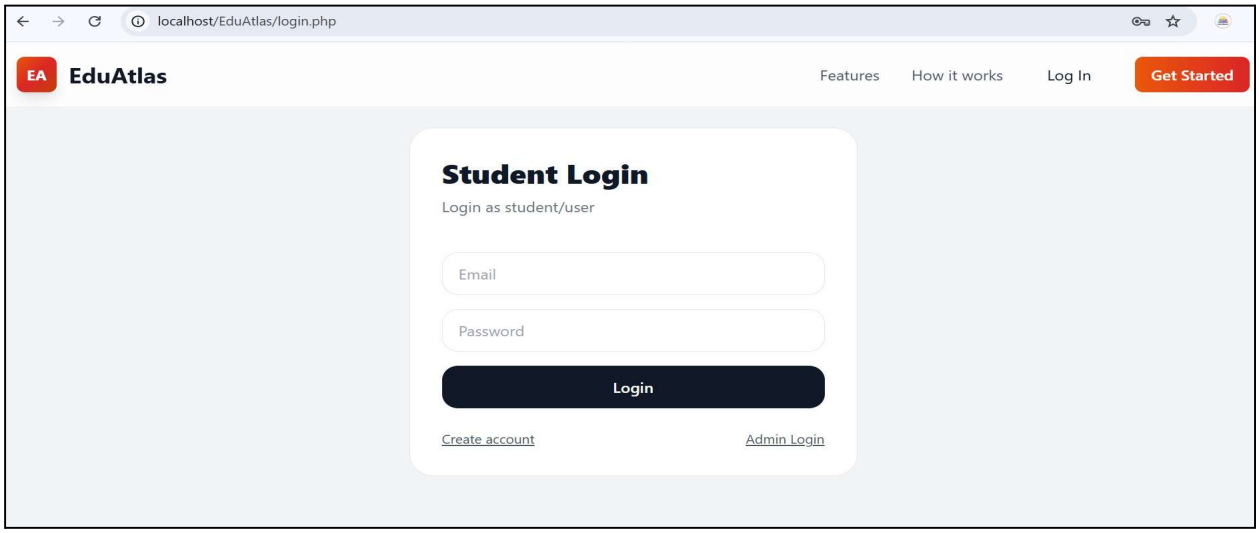

Figure 03 : Student Login of **EduAtlas.**

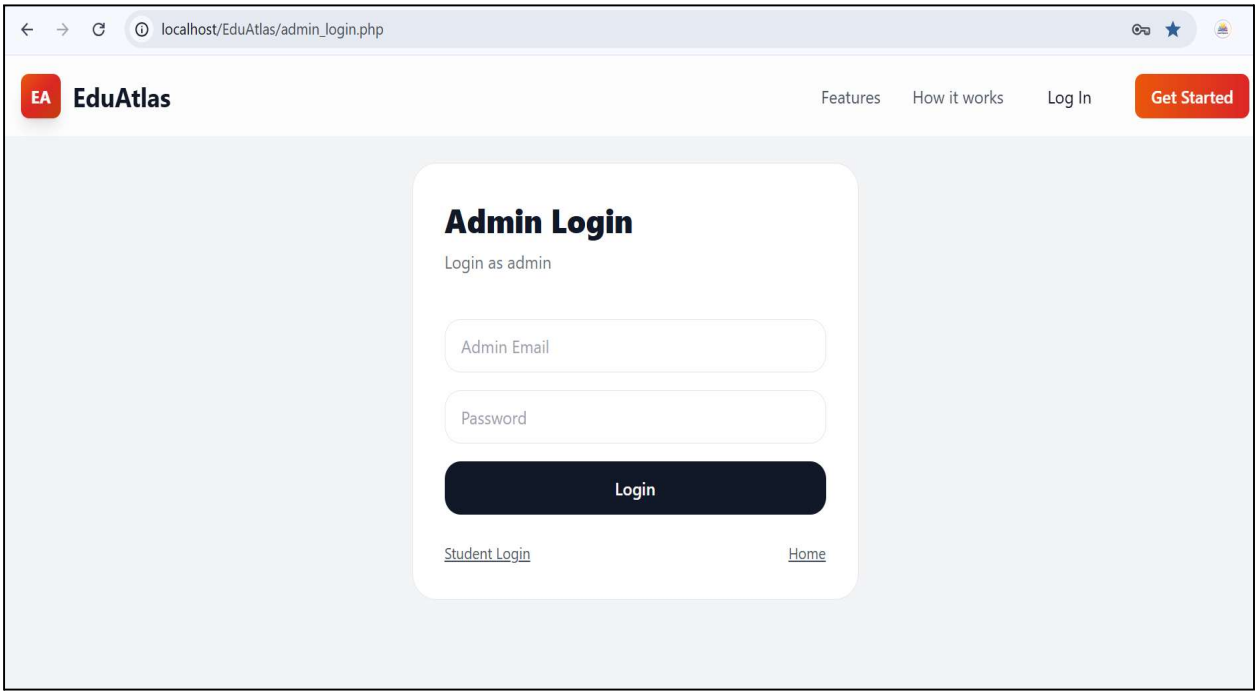

Figure 04 : Admin Login of **EduAtlas.**

6. ANALYSIS AND DISCUSSION :

In this lab, I started the backend implementation of my project **EduAtlas**, which is an admission assistance website for students. I successfully created the database **eduatlas** and connected it with PHP using PDO.

I implemented the basic server-side functionalities such as:

- Student login using session management.
- Admin login with role-based access.
- Creating a structured navigation system (navbar).

Also, implementation of some additional features is currently ongoing.

Through this implementation, I understood how server-side scripting works in a real web application. I learned how PHP interacts with MySQL to store and retrieve data. I also learned how session helps control user authentication and page access.

At the beginning, I faced some issues like database connection errors and query mistakes. After debugging and checking configuration settings, I was able to solve those problems.

Overall, this lab helped me understand:

- Database-driven web application development
- Secure login system implementation
- Role-based system design

This lab was an important step in starting the full implementation of my project.

7. SUMMARY :

The initial implementation of the EduAtlas admission assistance website was successfully completed. The database **eduatlas** was created and connected with PHP using PDO and basic server-side functionalities such as student and admin login, session management and dynamic display of institutions and resources were implemented.

This lab helped me understand database creation, PHP–MySQL connection,

server-side processing and role-based authentication. It provided a strong foundation for completing the full development of the project in the next stages.